Ethical Case Study Analysis: The Internet of Things and Information Professionals

In the modern world, the widespread integration of the internet has made all aspects of life more connected than ever. Internet connectivity extends beyond computers and smart phones to a variety of other devices. These devices are referred to as The Internet of Things (IoT), being a network of physical devices which are designed with sensors, software, and connectivity that allow them to collect and exchange data. These devices include everyday objects such as home appliances, cars, wearable health trackers, and more. While these devices and the mass amounts of data they collect can improve quality of life, they also raise many ethical concerns. This paper will examine the ethical and policy implications of IoT, as well as how IoT is relevant to me in my future career as an information professional.

The expansion of IoT is significantly changing the landscape of information connectivity in daily life. As more devices become internet-connected, digital systems increasingly operate in the background and collect data. According to Forward Thinking, estimates predicted that "for every person there will be 6.6 objects connected to the internet" (Fw:Thinking, 2013, 0:32). This level of connectivity growth has been made possible by technological developments like IPv6 networking standards, which create "more potential [internet] addresses than there are atoms on the surface of the earth" (Fw:Thinking, 2013, 2:28). As a result, a complex system of tools and sensors enabled for mass, interconnected data collection has emerged, and a major part of this data consists of personal information. In a 2025 article for Infophilia, Anita Coleman writes that a massive component of data collected by IoT devices is "IoB," or "Internet of Behavior [and] Bodies" (Coleman, 2025). IoB devices are intended to capture and monitor user habits, biomarkers, and other health related information, which can help to detect and prevent destabilizing medical events (Coleman, 2025).

However, it follows that IoT devices which capture highly sensitive personal information raise ethical concerns. Devices that track biometric data and behavioral patterns can create detailed digital profiles of individuals. Collecting and storing this information creates new vulnerabilities if the data is misused or exposed. As Coleman explains, “[w]hen our behavioral data ... becomes vulnerable, so do we. And once that data is collected ... it lives somewhere, potentially vulnerable to future breaches if not protected” (Coleman, 2025). Data collected by IoT devices like fitness trackers or biometric monitors can be stored indefinitely, and could be exposed even years later through security breaches. But security breaches are not the only problem. Companies also sell the data they collect from users. This risks private information going to institutions unknown to the original user, which can then be used in unknown and uncontrolled ways. Coleman writes that “[this] represents a fundamental shift in how capitalism extracts value—not just from your labor or your purchases, but from the very pattern of your existence ... transformed into behavioral surplus for prediction markets you'll never directly benefit from” (Coleman, 2025). Thus, not only is the information collected far beyond user control, but the user also receives no benefit or compensation from this practice. Considering that the spread of user data could lead to risks like hacking, blackmailing, discrimination, and more, there is a growing need to protect people whose data is harvested.

Therefore, the risks associated with IoT devices raise important questions about legal policy, responsibility, and ownership of data. Given that one of the most fundamental issues is determining who actually owns and controls the data generated by IoT devices, laws will need to keep pace with these advancing technologies. To address concerns like these, author Jack Karsten argues that policymakers must develop frameworks that clearly define “who owns [the] data; who can collect, store, mine and use it; and what penalties will be enforced if entities [act]

outside of [these] framework[s]" (Karsten, 2016). Without appropriate regulations, companies could continue to collect and profit off of personal data with limited accountability. Transparency is also an essential ethical feature when organizations collect information through IoT devices. Authors Singer and Perry note that "[c]lear communication is particularly important with respect to the passive collection of [information] or information that is not immediately evident to consumers" (Singer and Perry, 2015, p. 24). The authors also note that companies often attempt to address privacy concerns by aggregating data from many users and removing some personally identifiable information, but this practice still leaves questions about whether users can truly consent to what happens to their data (Singer and Perry, 2015, p. 26). With these factors in mind, it seems evident that governments should collaborate internationally to establish unified data privacy laws. These laws should mandate that users can opt out of data collection, storage, and sales, as well as entitlement to seeing what data is stored about them. The laws should also specify penalties for companies that improperly store data, are subject to breaches due to negligence, or retain unnecessary personally identifiable information. These penalties could include fines and orders to cease data collection and distribution.

The evolving landscape and ethical challenges associated with IoT are also directly relevant to my future career as an information professional. As IoT technologies continue to generate large quantities of personal and behavioral data, professionals who manage information systems will be an essential part of making sure this data is handled properly. Coleman notes the importance of the role of information professionals, stating that "[d]ifferent stakeholders have unique roles to play ... Information professionals must develop and advocate for transparent data stewardship practices" (Coleman, 2025). This means that the responsibility for implementing ethical data management strategies like minimizing unnecessary data collection, ensuring data

anonymity, and using secure data storage practices begins with information professionals. Further, information professionals can also create systems that allow individuals to opt out of data collection or request deletion of their personal data. By promoting transparency, individual privacy protection, and responsible data handling, information professionals can control the future trajectory of IoT safety. This will be especially important in the future as technologies continue to evolve, such as in the case of advancements in quantum computing technology and its implications for cybersecurity (Coleman, 2025).

In conclusion, the many ethical and policy implications of IoT remain relevant, especially to me in my future career as an information professional. While there are practical benefits of IoT devices, such as tracking personal health and fitness related information, there are also ethical concerns surrounding data collection and utilization that cannot be ignored. As both internet infrastructure and technologies enabling data collection continually integrate into modern life, individual users should have a reasonable amount of control over how their data is collected, stored, and distributed. When ethical operational and legal policies like safe data storage and data anonymization are used, information professionals can limit the severity of risks to users such as unauthorized data access, breaches, hacking, discrimination, and more. This is especially important because much of the time, individual users have more to lose than gain in the event their data becomes compromised. Laws should continually evolve and expand to protect individual privacies by penalizing companies and institutions who put user data at risk. If an international standard is established for data protection, along with maintaining individual user rights to one's data, then data can be ethically collected and managed. However, even with efforts taken to protect data, different kinds of risk cannot be fully eliminated, especially as

technologies continue to evolve into the future. Therefore, data collection should be limited to only what is essential and in users' best interest.

References

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